

Total No. of Questions : 9]

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S.E. (Electronics /E&TC/Electronics & Comp. Eng./Electronics

Engg.(VLSI Design & Tech.)/Electronics & Comm.(A.C.T.))

ENGINEERING MATHEMATICS - III

(2019 Pattern) (Semester - III) (207005)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Q.1 is compulsory.
- 2) Attempt Q.2 or Q.3, Q.4 or Q.5, Q.6 or Q.7, Q.8 or Q.9.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicate full marks.
- 5) Use of electronic pocket calculator is allowed.
- 6) Assume suitable data, if necessary.
- 7) Write numerical calculations correct upto four decimal places.

Q1) Write the correct option for the following multiple choice questions.

i) If $\phi = x^2 - y^2 + 2z^2$ then $\nabla\phi$ at the point (1, 2, 3) is _____ [2]

- | | |
|--------------------|--------------------|
| a) $2i - 4j - 12k$ | b) $2i - 4j + 12k$ |
| c) $2i + 4j + 12k$ | d) $2i + 4j - 12k$ |

ii) If $f(x) = x^2$, $h = 2$ then $\Delta^2 f(x)$ is given by [2]

- | | |
|------|-------|
| a) 6 | b) 12 |
| c) 4 | d) 8 |

iii) If $f(z) = \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)^2 (z-2)}$ then residue of $f(z)$ at $z = 2$ is _____ [2]

- | | |
|------|----------|
| a) 0 | b) -1 |
| c) 1 | d) π |

P.T.O.

OR

Q3) a) Given $\sin 45^\circ = 0.7071$, $\sin 50^\circ = 0.7660$, $\sin 55^\circ = 0.8192$, $\sin 60^\circ = 0.8660$, Find $\sin 58^\circ$ using Newton's backward difference formula. [5]

b) Given $\frac{dy}{dx} = x^2 - y$, $y(0) = 1$, find $y(0.1)$ using Runge-Kutta method of fourth order (Take $h = 0.1$) [5]

c) Use Trapezoidal rule to evaluate $I = \int_{-3}^3 x^4 dx$ using six equal subintervals. [5]

Q4) a) Find the directional derivative of $\phi = xy + yz + zx$ at $(1, 1, 1)$ along the vector $\vec{i} + 2\vec{j} + 2\vec{k}$ [5]

b) Show that the vector field

$\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational. Find scalar potential function ϕ such that $\vec{F} = \nabla\phi$ [5]

c) Find angle between tangents to the curve $x = t$, $y = t^2$, $z = t^3$ at $t = 1$ and $t = -1$. [5]

OR

Q5) a) Find the directional derivative of $\phi = xyz$ at $(1, 2, -1)$ in the direction normal to the surface $x \log z - y^2 = -4$ at $(-1, 2, 1)$ [5]

b) Prove that $\nabla^2 \left[\nabla \cdot \frac{\vec{r}}{r^2} \right] = \frac{2}{r^4}$ [5]

c) The position vector of a particle at time t is

$\vec{r} = \cos(t-1)\vec{i} + \sin h(t-1)\vec{j} + mt^3\vec{k}$. Find the condition on m . So that at any time $t = 1$ the acceleration is normal to the position vector. [5]

Q6) a) Apply Green's Theorem to evaluate $\int_C (3y dx + 2x dy)$ where C is boundary of $0 \leq x \leq \pi$, $0 \leq y \leq \sin x$ [5]

b) Using Gauss-divergence theorem, prove that

$$\iiint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot d\vec{s} = \iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dV$$
 [5]

- c) Using Stoke's theorem, evaluate : [5]

$$\int_C [3(x-y)dx + 2xzdy + xydz]$$

Where C is the curve of intersection of paraboloid $x^2 + y^2 = 2z$ and the plane $z = 2$

OR

- Q7) a) Evaluate $\int_C \vec{F} \cdot d\vec{r}$, $\vec{F} = (2x + y^2)\vec{i} + (3y - 4x)\vec{j}$ where C is the parabolic arc $y = x^2$ joining (0, 0) & (1, 1) [5]

- b) Using Gauss-Divergence Theorem, evaluate $\iiint_S \vec{F} \cdot \hat{n} ds$ for $\vec{F} = 4xz\vec{i} - y^2\vec{j} + yz\vec{k}$ where S is the surface of the cube bounded by the planes $x = 0, x = 2, y = 0, y = 2, z = 0, z = 2$ [5]

- c) Evaluate $\iint_S (\nabla \times \vec{F}) \cdot d\vec{s}$ for $\vec{F} = y\vec{i} + z\vec{j} + x\vec{k}$ where S is the surface of the paraboloid $z = 1 - x^2 - y^2, z \geq 0$ [5]

- Q8) a) If $u = x^4 - 6x^2y^2 + y^4$, find V such that $f(z) = u + iv$ is analytic function. Express $f(z)$ in terms of z. [5]

- b) Use Cauchy's integral formula to evaluate $\oint_C \frac{e^z}{z+2} dz$ where C is the circle $|z + 2| = 2$ [5]

- c) Find the bilinear transformation which maps the points 0, -1, i of the z-plane onto the points 2, $\infty, \frac{1}{2}(5 + i)$ of the W-plane. [5]

OR

- Q9) a) Show that analytic function $f(z)$ with constant amplitude is constant. [5]

- b) Apply residue theorem to evaluate $\oint_C \frac{4z^2 + z}{z^2 - 1} dz$ where C is the contour $|z| = \frac{1}{2}$ [5]

- c) Show that the transformation $W = z + \frac{1}{z} - 2i$ maps the circle $|z| = 2$ into an ellipse. Find centre, semi-major and semi-minor axes of ellipse. [5]

